BASE CHANGE OF THE TAME SYMBOL ALONG A FLAT LOCAL HOMOMORPHISM OF 1-DIMENSIONAL NOETHERIAN LOCAL RINGS

THE STACKS PROJECT AUTHORS

Editorial note. Counted proposition: Tag 0EPG, chow.tex lines 999-1008 with its own proof at lines 1010-1086 (statement immediately followed by its proof; no non-adjacent pairing).

Editorial note. The context blocks reproduce the author's definition of the tame symbol with its defining formula (Tag 0EAQ) and the statement of the factorization lemma Tag 0EAG; the latter is also counted as a separate task of this batch, so its proof is not duplicated here.

Editorial note. The displayed identity labelled equation-to-prove at chow.tex lines 1055-1061 is created inside the proof and is not an external reference.

Editorial note. External references are navigation pointers to stable Stacks tags; the determinant computation that uses them is retained in full in the counted proof.

Allowed standard prerequisites (editorial).

- The tame symbol of a 1-dimensional Noetherian local ring. Stacks tag 0EAQ (retained context, chow.tex lines 759-827). Fixes the objects partial_A(a_1, a_2) in kappa(m) and the local-factorization recipe that the statement compares with partial_{A'}(a_1, a_2).
- Local factorization after a finite extension. Stacks tag 0EAG (retained context, chow.tex lines 641-655; counted as a separate task of this batch). Supplies the finite extension A contained in B and the factorizations a_i = u_{i,j} pi_j^{e_{i,j}} at the maximal ideals of B, used by the author at lines 1013-1020.
- Determinants of endomorphisms of finite-length modules, their multiplicativity and their flat base change. More on Algebra, Section 0GSX and Lemmas 0GSZ, 0GT0, 0GSY. Rewrites Norm(u)^(m_j) as det_kappa(u : M to M) for M = B_j/pi_j B_j, moves it to det_{kappa'}(u : M tensor A' to M tensor A') and splits it over the maximal ideals of B' = A' tensor_A B (author's lines 1062-1085).
- Identification of the residue field extension through the finite flat extension. Stacks tag 02M0 (Algebra, Lemma 02M0), used in the retained context. Underlies the length and degree bookkeeping m_j, m'_{j,l} that the comparison of the two displayed formulae relies on.

Source reference. Original reference 02C5: sources/stacks-chow/tags.txt, source lines 3072–3072. The total ring of fractions Q(A) is introduced at chow.tex lines 763-765 by a pointer to this example; the counted proof never uses Q(A), only nonzerodivisors of A.

Date: stacks-project:chow.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original section 0GSX: sources/stacks-chow/tags.txt, source lines 20947–20947. Navigation pointer to the construction of determinants of endomorphisms of finite-length modules, invoked at chow.tex line 1064; the author's determinant comparison itself is retained in full.

Source reference. Original lemma 0GSZ: sources/stacks-chow/tags.txt, source lines 20949–20949. Identifies the norm power with a determinant, cited at chow.tex lines 1066 and 1083; a standard input to the retained computation, not a substitute for it.

Source reference. Original lemma 0GT0: sources/stacks-chow/tags.txt, source lines 20950–20950. Flat base change of the determinant construction, cited at chow.tex line 1072; the flatness hypothesis and its use are stated by the author, and the comparison remains the counted content.

Source reference. Original lemma 0GSY: sources/stacks-chow/tags.txt, source lines 20948–20948. Multiplicativity of the determinant construction in short exact sequences, cited at chow.tex line 1085 to conclude the displayed identity.

Lemma 0.1. *Let $(A, \mathfrak{m})$ be a Noetherian local ring of dimension 1. Let $r \geq 2$ and let $a_1, \ldots, a_r \in A$ be nonzerodivisors not all units. Then there exist*

1. *a finite ring extension $A \subset B$ with B/A annihilated by a power of $\mathfrak{m}$,*
2. *for each maximal ideal $\mathfrak{m}_j \subset B$ a nonzerodivisor $\pi_j \in B_j = B_{\mathfrak{m}_j}$, and*
3. *factorizations $a_i = u_{i,j}\pi_j^{e_{i,j}}$ in B_j with $u_{i,j} \in B_j$ units and $e_{i,j} \geq 0$.*

1. Tame symbols

Consider a Noetherian local ring $(A, \mathfrak{m})$ of dimension 1. We denote $Q(A)$ the total ring of fractions of A, see Algebra, Example 02C5. The *tame symbol* will be a map

$$\partial_A(-,-) : Q(A)^* \times Q(A)^* \longrightarrow \kappa(\mathfrak{m})^*$$

satisfying the following properties:

1. $\partial_A(f, gh) = \partial_A(f, g)\partial_A(f, h)$ for $f, g, h \in Q(A)^*$,
2. $\partial_A(f, g)\partial_A(g, f) = 1$ for $f, g \in Q(A)^*$,
3. $\partial_A(f, 1 - f) = 1$ for $f \in Q(A)^*$ such that $1 - f \in Q(A)^*$,
4. $\partial_A(aa', b) = \partial_A(a, b)\partial_A(a', b)$ and $\partial_A(a, bb') = \partial_A(a, b)\partial_A(a, b')$ for $a, a', b, b' \in A$ nonzerodivisors,
5. $\partial_A(b, b) = (-1)^m$ with $m = \text{length}_A(A/bA)$ for $b \in A$ a nonzerodivisor,
6. $\partial_A(u, b) = u^m \bmod \mathfrak{m}$ with $m = \text{length}_A(A/bA)$ for $u \in A$ a unit and $b \in A$ a nonzerodivisor, and
7. $\partial_A(a, b - a)\partial_A(b, b) = \partial_A(b, b - a)\partial_A(a, b)$ for $a, b \in A$ such that $a, b, b - a$ are nonzerodivisors.

Since it is easier to work with elements of A we will often think of ∂_A as a map defined on pairs of nonzerodivisors of A satisfying (4), (5), (6), (7). It is an exercise to see that setting

$$\partial_A(\frac{a}{b}, \frac{c}{d}) = \partial_A(a, c)\partial_A(a, d)^{-1}\partial_A(b, c)^{-1}\partial_A(b, d)$$

we get a well defined map $Q(A)^* \times Q(A)^* \to \kappa(\mathfrak{m})^*$ satisfying (1), (2), (3) as well as the other properties.

We do not claim there is a unique map with these properties. Instead, we will give a recipe for constructing such a map. Namely, given $a_1, a_2 \in A$ nonzerodivisors, we choose a ring extension $A \subset B$ and local factorizations as in Lemma 0.1. Then we define

$$\partial_A(a_1, a_2) = \prod\nolimits_j \text{Norm}_{\kappa(\mathfrak{m}_j)/\kappa(\mathfrak{m})}((-1)^{e_{1,j}e_{2,j}} u_{1,j}^{e_{2,j}} u_{2,j}^{-e_{1,j}} \bmod \mathfrak{m}_j)^{m_j} \tag{1.0.1}$$

where $m_j = \text{length}_{B_j}(B_j/\pi_j B_j)$ and the product is taken over the maximal ideals $\mathfrak{m}_1, \ldots, \mathfrak{m}_r$ of B.

Lemma 1.1. *Let $(A, \mathfrak{m}, \kappa) \to (A', \mathfrak{m}', \kappa')$ be a local homomorphism of Noetherian local rings. Assume $A \to A'$ is flat and $\dim(A) = \dim(A') = 1$. Set $m = length_{A'}(A'/\mathfrak{m}A')$. For $a_1, a_2 \in A$ nonzerodivisors $\partial_A(a_1, a_2)^m$ maps to $\partial_{A'}(a_1, a_2)$ via $\kappa \to \kappa'$.*

Proof. If a_1, a_2 are both units, then $\partial_A(a_1, a_2) = 1$ and $\partial_{A'}(a_1, a_2) = 1$ and the result is true. If not, then we can choose a ring extension $A \subset B$ and local factorizations as in Lemma 0.1. Denote $\mathfrak{m}_1, \ldots, \mathfrak{m}_m$ be the maximal ideals of B. Let $\mathfrak{m}_1, \ldots, \mathfrak{m}_m$ be the maximal ideals of B with residue fields $\kappa_1, \ldots, \kappa_m$. For each $j \in \{1, \ldots, m\}$ denote $\pi_j \in B_j = B_{\mathfrak{m}_j}$ a nonzerodivisor such that we have factorizations $a_i = u_{i,j}\pi_j^{e_{i,j}}$ as in the lemma. By definition we have

$$\partial_A(a_1, a_2) = \prod\nolimits_j \text{Norm}_{\kappa_j/\kappa}((-1)^{e_{1,j}e_{2,j}} u_{1,j}^{e_{2,j}} u_{2,j}^{-e_{1,j}} \bmod \mathfrak{m}_j)^{m_j}$$

where $m_j = \text{length}_{B_j}(B_j/\pi_j B_j)$.

Set $B' = A' \otimes_A B$. Since A' is flat over A we see that $A' \subset B'$ is a ring extension with B'/A' annihilated by a power of $\mathfrak{m}'$. Let

$$\mathfrak{m}'_{j,l}, \quad l = 1, \ldots, n_j$$

be the maximal ideals of B' lying over $\mathfrak{m}_j$. Denote $\kappa'_{j,l}$ the residue field of $\mathfrak{m}'_{j,l}$. Denote $B'_{j,l}$ the localization of B' at $\mathfrak{m}'_{j,l}$. As factorizations of a_1 and a_2 in $B'_{j,l}$ we use the image of the factorizations $a_i = u_{i,j}\pi_j^{e_{i,j}}$ given to us in B_j. By definition we have

$$\partial_{A'}(a_1, a_2) = \prod\nolimits_{j,l} \text{Norm}_{\kappa'_{j,l}/\kappa'}((-1)^{e_{1,j}e_{2,j}} u_{1,j}^{e_{2,j}} u_{2,j}^{-e_{1,j}} \bmod \mathfrak{m}'_{j,l})^{m'_{j,l}}$$

where $m'_{j,l} = \text{length}_{B'_{j,l}}(B'_{j,l}/\pi_j B'_{j,l})$.

Comparing the formulae above we see that it suffices to show that for each j and for any unit $u \in B_j$ we have

$$\big(\text{Norm}_{\kappa_j/\kappa}(u \bmod \mathfrak{m}_j)^{m_j}\big)^m = \prod\nolimits_l \text{Norm}_{\kappa'_{j,l}/\kappa'}(u \bmod \mathfrak{m}'_{j,l})^{m'_{j,l}} \tag{1.1.1}$$

in κ'. We are going to use the construction of determinants of endomorphisms of finite length modules in More on Algebra, Section 0GSX to prove this. Set $M = B_j/\pi_j B_j$. By More on Algebra, Lemma 0GSZ we have

$$\text{Norm}_{\kappa_j/\kappa}(u \bmod \mathfrak{m}_j)^{m_j} = \det_\kappa(u : M \to M)$$

Thus, by More on Algebra, Lemma 0GT0, the left hand side of (1.1.1) is equal to $\det_{\kappa'}(u : M \otimes_A A' \to M \otimes_A A')$. We have an isomorphism

$$M \otimes_A A' = (B_j/\pi_j B_j) \otimes_A A' = \bigoplus\nolimits_l B'_{j,l}/\pi_j B'_{j,l}$$

of A'-modules. Setting $M'_l = B'_{j,l}/\pi_j B'_{j,l}$ we see that $\text{Norm}_{\kappa'_{j,l}/\kappa'}(u \bmod \mathfrak{m}'_{j,l})^{m'_{j,l}} = \det_{\kappa'}(u_j : M'_l \to M'_l)$ by More on Algebra, Lemma 0GSZ again. Hence (1.1.1) holds by multiplicativity of the determinant construction, see More on Algebra, Lemma 0GSY. □